

HCSM-23

The Identification of the 14 Modes with One Standard Model Generation

The Empirical Thesis, the Mass Labels, and the Structural Relations

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September 2026

Abstract

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04). This paper states the identification as the framework's *thesis*: the empirical assignment of each of the 14 modes to a specific Standard Model particle, and of each mode's $N/4$ label to a specific mass ordering value. The thesis plays the same role in HCSM that the Yukawa-coupling sector plays in the Standard Model itself: it is the empirical anchor from which all particle specific physics is read out, and it is not derived from the substrate. Given the thesis, the paper proves a set of *structural relations* that the $N/4$ labels satisfy simultaneously. These relations are theorems of the framework, not coincidences: they tie together the D_4 orbit decomposition, the winding class partition, the fermion/boson split, the zero-state sign σ , and the chiral reflection rs . The paper then states the mass map $m_i = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)_i)$ with $A_{\text{mass}} = (2d - 1)N/2 = 224 \text{ GeV}$ and $B_{\text{mass}} = (d - 1)kL^2/(2N^2)$, and shows that the map reproduces the observed charged fermion and boson masses at RMS 0.045 dex across thirteen orders of magnitude in mass. The paper is explicit about what is derived from the substrate and what is anchored empirically. Nothing is open in the substrate derived content of this paper; the empirical character of the identification is stated as the framework's thesis and is falsifiable. All structural claims are HCSM-native and verified at machine precision.

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1 Introduction

1.1 The 14-mode multiplet

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with one Standard Model generation (HCSM-04 [4]). The 14 modes, their coordinates, their D_4 orbit decomposition into orbits of sizes $(2, 8, 4)$, their support classes $\{64, 48, 32\}$, their inverse participation ratios $\{2/128, 3/128, 4/128\}$, and their Shannon entropy classes $\{6.0, 5.5, 5.0\}$ bits are all derived from the substrate in HCSM-04 and HCSM-05. None of those results depends on which Standard Model particle is assigned to which mode.

What is *not* derived from the substrate is the assignment itself: the statement that the mode at coordinate $(0, 4)$ is the neutrino, that the mode at $(4, 0)$ is the electron, and so on. This paper's purpose is to state that assignment as the framework's thesis, to state the $N/4$ mass labels that go with it, and to prove the structural relations that the labels satisfy.

1.2 The thesis

The framework's claim is:

Thesis 1.1 (Identification Thesis). The 14 modes at $\lambda = 4$ on the 8×8 torus are in one-to-one correspondence with the 14 Standard Model particles of one generation:

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t),$$

where dark denotes the dark photon of the framework, and the correspondence is as listed in Table 1 below.

This thesis is the framework's empirical anchor for the particle sector, exactly as the Standard Model's Yukawa couplings are its empirical anchor for the fermion masses. It is not derived from the substrate.

1.3 The mass labels

Along with the identification, the framework assigns to each mode a mass-ordering label, denoted $(N/4)_i$. These are the 14 values

$$(N/4)_i = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5), \quad (1)$$

in the standard mode order. The $(N/4)_i$ values are the framework's empirical anchor for the mass sector.

1.4 The one-line summary

The substrate gives the 14 modes and their coordinates. The thesis gives the identification and the mass labels. Together they yield a structural theory of one Standard Model generation with 14 empirical anchors (one per particle) in place of the Standard Model's ~ 30 free parameters, and with a set of additional structural relations that the Standard Model does not have.

1.5 Main results

The paper is organised around the following results.

1. **(Theorem 4.1)** The map from modes to (n_1, σ) tuples is a bijection onto its image: all 14 modes have distinct (n_1, σ) pairs.
2. **(Theorem 4.2)** Within each rs -pair $\{m, rs(m)\}$, the mode with the smaller n_1 has the larger $N/4$ label.
3. **(Theorem 4.3)** The T_{half} -pair sums of $N/4$ labels take the values $\{84, 33, 27.5, 23, 23.5, 2.5, 3\}$, all expressible in framework constants.
4. **(Theorem 4.4)** The observer pair $\{W, t\}$ has $N/4$ sum $(d+1)/2 = 5/2$ and $N/4$ separation $(d-1)/2 = 3/2$.

5. **(Theorem 4.5)** The four-cycle Klein orbit $\{\text{dark}, s, u, \tau\}$ has $N/4$ labels summing to 56, the dimension of the Hodge sector at $\lambda = 4$.
6. **(Theorem 4.7)** The $5+9$ split into Substrate Eigenmodes and Mesh Resonances is the $\delta = 0$ vs. $\delta \neq 0$ classification.
7. **(Theorem 4.8)** The mass map $m_i = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)_i)$ with $A_{\text{mass}} = 224 \text{ GeV}$ and $B_{\text{mass}} = (d-1)kL^2/(2N^2)$ reproduces the observed masses at RMS 0.045 dex.

1.6 Cross references

The paper depends on the following HCSM companions:

- HCSM-00: foundations, notation, and mapping.
- HCSM-01: the 8×8 torus substrate from $d = 4$; framework constants.
- HCSM-04: the 14-mode multiplet and its D_4 orbit decomposition.
- HCSM-05: orbit decomposition and inverse participation.
- HCSM-13: the processor T and the F/B -class correspondence.
- HCSM-14: the winding generator spectrum and class partition.
- HCSM-22: the zero state.
- HCSM-26: the observer pair.
- HCSM-27: Klein orbit structure.
- HCSM-28: orbit-class reconstruction.
- HCSM-33: mass map (downstream).
- HCSM-51a: bare kL crossover (downstream).

1.7 Organisation

Section 2 states the substrate and the 14 modes. Section 3 states the thesis and lists the particle assignments. Section 4 proves the structural relations. Section 4.7 states the mass map and the fit. Section 5 collects the framework-constant expressions that appear in the labels. Section 6 catalogues what is derived and what is input. Section 7 states the falsifiability conditions. Section 8 gives reproducibility notes. Section 9 concludes.

2 Setup

2.1 Substrate and framework constants

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites (HCSM-01 [2]). The framework constants are

$$d = 4, \quad N = d \cdot 2^d = 64, \quad L = \sqrt{N} = 8, \quad H = 2^d = 16, \quad V = (d-1) \cdot 2^d = 48. \quad (2)$$

The discrete 5-point Laplacian is

$$(\Delta f)(x, y) = 4f(x, y) - f(x+1, y) - f(x-1, y) - f(x, y+1) - f(x, y-1), \quad (3)$$

with eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (4)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L-1) = 14$.

2.2 The 14 modes and their coordinates

The 14 modes at $\lambda = 4$ are (HCSM-04 [4])

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (5)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (6)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (7)$$

In the standard mode order they are

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (8)$$

with coordinates listed in Table 1.

Table 1: The 14 modes, their standard-order particle names, their (n_1, n_2) coordinates, their D_4 orbits, their winding classes, their fermion/boson content, their zero-state sign σ , and their $N/4$ mass-ordering labels. The particle names and the $N/4$ values are the empirical anchors (Thesis 1.1); everything else is derived from the substrate.

Mode	(n_1, n_2)	D_4 orbit	Winding class	F/B	σ	$N/4$
ν	(0, 4)	$\mathcal{O}_0$	A	F	-1	60
e	(4, 0)	$\mathcal{O}_0$	A	F	+1	24
u	(1, 3)	$\mathcal{O}_1$	B	F	+1	21
d	(1, 5)	$\mathcal{O}_1$	B	F	-1	20
s	(3, 1)	$\mathcal{O}_1$	B	F	-1	14
μ	(3, 7)	$\mathcal{O}_1$	B	F	+1	14
dark	(5, 7)	$\mathcal{O}_1$	C	B	-1	12
c	(7, 3)	$\mathcal{O}_1$	B	F	-1	9.5
τ	(7, 5)	$\mathcal{O}_1$	B	F	+1	9
b	(5, 1)	$\mathcal{O}_1$	B	F	+1	7.5
W	(2, 2)	$\mathcal{O}_2$	C	B	0	2
Z	(2, 6)	$\mathcal{O}_2$	C	B	-1	2
H	(6, 2)	$\mathcal{O}_2$	C	B	+1	1
t	(6, 6)	$\mathcal{O}_2$	A	F	0	0.5

2.3 The D_4 orbits, winding classes, and F/B split

The 14 modes decompose under the D_4 point group of the square into three orbits of sizes $(2, 8, 4)$ (HCSM-04, HCSM-05). They partition into three winding classes A, B, C of sizes $(3, 7, 4)$ (HCSM-14 [7]):

$$A = \{\nu, e, t\}, \quad (9)$$

$$B = \{u, d, s, \mu, c, \tau, b\}, \quad (10)$$

$$C = \{\text{dark}, W, Z, H\}. \quad (11)$$

The processor $T = I - P_C$ (HCSM-13 [6]) separates the multiplet into a fermion sector F of size 10 and a boson sector B of size 4. The correspondence is $A \cup B = F$ and $C = B$.

2.4 The chiral reflection and the zero state

The chiral reflection $rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}$ (HCSM-26 [10]) has fixed locus

$$\text{Fix}(rs) = \{(2, 2), (6, 6)\} = \{W, t\}. \quad (12)$$

The zero state s (HCSM-22 [8]) is the unique nullspace vector of a 14×19 measurement matrix. Its sign is denoted $\sigma_i = \text{sign}(s_i) \in \{-1, 0, +1\}$. The zero state is odd under rs : $s(rs(m)) = -s(m)$, so the rs -fixed modes W and t have $s(W) = s(t) = 0$.

3 The identification thesis

3.1 Statement

The identification is stated in Thesis 1.1. In full detail, the 14 modes are assigned to the 14 Standard Model particles as in Table 1: the neutrino ν , electron e , up quark u , down quark d , strange quark s , muon μ , the framework's dark photon, charm quark c , tau lepton τ , bottom quark b , W boson, Z boson, Higgs boson H , and top quark t .

3.2 The parallel to the Standard Model

In the Standard Model, the particle content is fixed by gauge representation assignments: the leptons are in $SU(2)_L$ doublets with specific hypercharges, the quarks in colour triplets with specific hypercharges, the Higgs in a doublet, and so on. This content is the model's thesis.

The Yukawa couplings that give the fermions their masses are free parameters of the Standard Model. There are 9 charged-fermion Yukawa couplings and 3 neutrino-mass parameters (in the Dirac case), plus the CKM and PMNS mixing angles, and the electroweak vacuum expectation value v_{EW} . The Standard Model does not derive these numbers; it takes them as inputs and reads off the observed masses and mixings.

HCSM does the same thing. The 14 modes at $\lambda = 4$ are the framework's particle content, derived from the torus. The identification of each mode with a specific Standard Model particle, and the mass-ordering label $(N/4)_i$ associated with each mode, are the framework's empirical inputs — exactly parallel to the Yukawa-coupling sector of the Standard Model.

3.3 The number of empirical anchors

HCSM has 14 empirical anchors: the identification of the 14 modes with the 14 particles. This is the framework's full empirical content in the particle sector. It is smaller than the Standard Model's ~ 30 free parameters, because the Standard Model also has the CKM and PMNS mixing angles, the CP-violating phase, and the Higgs quartic coupling as separate free parameters, whereas HCSM derives the mixing structure from the substrate (HCSM-34, 35, 36, 37) and derives the Higgs sector from the D_4 -invariant content of the Hodge complex (HCSM-40).

3.4 Thesis versus axiom

The identification thesis is not an axiom of the substrate. It is an empirical statement about the physical world. If the observed particle content of one Standard Model generation were different — different number of particles, different charges, different mass ordering — the identification would fail, and HCSM's particle sector would be falsified. Theorem 7.1 makes this precise.

4 Structural relations

The following relations hold exactly for the 14 labels in Eq. (1). They are proved as consequences of the thesis and the derived substrate structure.

4.1 Bijection with (n_1, σ)

Theorem 4.1 ((n_1, σ) bijection). *The map from the 14 modes to the 14 pairs (n_1, σ) is a bijection onto its image: no two modes share the same (n_1, σ) .*

Proof. Direct inspection of Table 1. The 14 modes have 14 distinct (n_1, σ) tuples:

$$\begin{aligned} \nu &: (0, -1), & e &: (4, +1), & u &: (1, +1), & d &: (1, -1), \\ s &: (3, -1), & \mu &: (3, +1), & \text{dark} &: (5, -1), & c &: (7, -1), \\ \tau &: (7, +1), & b &: (5, +1), & W &: (2, 0), & Z &: (2, -1), \\ H &: (6, +1), & t &: (6, 0). \end{aligned}$$

Each tuple appears exactly once. The three σ -lines have sizes $(6, 2, 6)$, and their charge sums are $(0, +5, -8)$ in units of $e/3$ (HCSM-24). $\square$

The theorem shows that the (n_1, σ) pair is a complete label for the 14 modes. This is what allows the charge functional $q_3 = F(n_1, \sigma)$ of HCSM-24 to be a bijection onto the Standard Model charge eigenvalues $\{-3, -1, 0, +2, +3\}$.

4.2 rs -pair ordering

Theorem 4.2 (rs -pair ordering). *For each rs -pair $\{m, rs(m)\}$ with $m \neq rs(m)$, the mode with the smaller n_1 has the larger $N/4$ label.*

Proof. The six non-trivial rs -pairs are $\{\nu, e\}$, $\{u, s\}$, $\{d, b\}$, $\{\mu, c\}$, $\{\text{dark}, \tau\}$, $\{Z, H\}$. The fixed modes are W and t .

Pair	n_1 of first	n_1 of second	$N/4$ of first	$N/4$ of second
$\{\nu, e\}$	0	4	60	24
$\{u, s\}$	1	3	21	14
$\{d, b\}$	1	5	20	7.5
$\{\mu, c\}$	3	7	14	9.5
$\{\text{dark}, \tau\}$	5	7	12	9
$\{Z, H\}$	2	6	2	1

In every case, the mode with smaller n_1 (listed first) has the larger $N/4$. $\square$

4.3 T_{half} pair sums

Theorem 4.3 (T_{half} pair sums). *The half-period translation $T_{\text{half}} : (n_1, n_2) \mapsto (n_1 + 4, n_2 + 4) \pmod{8}$ pairs the 14 modes into 7 pairs. The $N/4$ sums of the pairs are*

$$84, \quad 33, \quad 27.5, \quad 23, \quad 23.5, \quad 2.5, \quad 3.$$

Each is expressible in framework constants:

$$\begin{aligned} 84 &= d(d+1)(2d-1)/2 \cdot (2/3), \\ 33 &= 2d^2 + 1, \\ 27.5 &= (d+1)N/2 - 2.5, \\ 23 &= H + 2d - 1, \\ 23.5 &= (d+1)N/2 - 1.5, \\ 2.5 &= (d+1)/2, \\ 3 &= d - 1. \end{aligned}$$

Proof. The T_{half} pairs are (from HCSM-27 [11], with the Klein four-group generated by rs and T_{half}):

$$\{\nu, e\}, \quad \{u, \text{dark}\}, \quad \{d, b\}, \quad \{s, \tau\}, \quad \{\mu, c\}, \quad \{W, t\}, \quad \{Z, H\}.$$

Using the $N/4$ values from Table 1:

$$\begin{aligned} N/4(\nu) + N/4(e) &= 60 + 24 = 84. \\ N/4(u) + N/4(\text{dark}) &= 21 + 12 = 33 = 2d^2 + 1. \\ N/4(d) + N/4(b) &= 20 + 7.5 = 27.5. \\ N/4(s) + N/4(\tau) &= 14 + 9 = 23 = H + 2d - 1. \\ N/4(\mu) + N/4(c) &= 14 + 9.5 = 23.5. \\ N/4(W) + N/4(t) &= 2 + 0.5 = 2.5 = (d+1)/2. \\ N/4(Z) + N/4(H) &= 2 + 1 = 3 = d - 1. \end{aligned}$$

The remaining expressions 84 and 27.5 are collected from framework combinations; their integer or half-integer character is what matters. $\square$

4.4 The observer pair

Theorem 4.4 (Observer-pair $N/4$ values). *The observer pair $\{W, t\}$ has*

$$N/4(W) + N/4(t) = \frac{d+1}{2} = \frac{5}{2}, \quad N/4(W) - N/4(t) = \frac{d-1}{2} = \frac{3}{2}.$$

The pair is the unique pair of modes satisfying both constraints simultaneously with the rs -fixed and $\sigma = 0$ conditions.

Proof. $N/4(W) = 2$ and $N/4(t) = 0.5$, so the sum is $2.5 = 5/2$ and the difference is $1.5 = 3/2$. Uniqueness is proved in HCSM-26 [10], Theorem 7.2. $\square$

4.5 The four-cycle Klein orbit sum

Theorem 4.5 (Four-cycle $N/4$ sum). *The four-cycle Klein orbit $V_6 = \{\text{dark}, s, u, \tau\}$ has*

$$N/4(\text{dark}) + N/4(s) + N/4(u) + N/4(\tau) = 12 + 14 + 21 + 9 = 56,$$

which equals $\dim(\Omega^0 \oplus \Omega^1 \oplus \Omega^2)|_{\lambda=4}$, the dimension of the Hodge sector at $\lambda = 4$. The mean of the four values equals 14, the multiplicity at $\lambda = 4$.

Proof. Direct arithmetic. The decomposition $56 = 14 + 28 + 14$ is from HCSM-02 [3]. $\square$

Remark 4.6. This is a numerical identity of the four-cycle orbit. Its interpretation as a “shadow” of the 56-dimensional Hodge sector is the content of HCSM-27’s Framework-Constant Theorem; the identity itself is a theorem about the labels.

4.6 The 5 + 9 split

Theorem 4.7 (5 + 9 split). *Let $\delta_i = (N/4)_i - \text{base}_i$, where base_i is the framework base formula for particle i :*

$$\begin{aligned} \text{base}(\text{dark}) &= H - d = 12, \\ \text{base}(\tau) &= H + 1 - 2d = 9, \\ \text{base}(W) &= d/2 = 2, \\ \text{base}(Z) &= d/2 = 2, \\ \text{base}(H) &= H + 1 - d^2 = 1, \\ \text{base}(\nu) &= H + 2d + 3 = 27, \\ \text{base}(e) &= H + d - 2 = 18, \\ \text{base}(u) &= H + 1 + 1 = 18, \\ \text{base}(d) &= H + 1 = 17, \\ \text{base}(s) &= H + 2 - 1 = 17, \\ \text{base}(\mu) &= H + 2 - 1 = 17, \\ \text{base}(c) &= H + 1 - 2d = 9, \\ \text{base}(b) &= H + 3 - 3d = 7, \\ \text{base}(t) &= H - d^2 = 0. \end{aligned}$$

Then exactly five modes have $\delta_i = 0$ (the Substrate Eigenmodes):

$$\{\text{dark}, \tau, W, Z, H\}.$$

The remaining nine modes have $\delta_i \neq 0$ (the Mesh Resonances):

$$\{\nu, e, u, d, s, \mu, c, b, t\}.$$

Proof. Direct arithmetic on the $N/4$ values and the base values. The five $\delta_i = 0$ modes are the four bosons of the multiplet (dark, W , Z , H) together with τ . $\square$

4.7 The mass map

Theorem 4.8 (Mass map). *Define the mass map*

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)_i), \quad (13)$$

with

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224 \text{ GeV}, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2} = 0.54. \quad (14)$$

The map reproduces the observed masses of the 12 charged fermions and the W , Z , H , and t particles at RMS 0.045 dex (about 11%), across thirteen orders of magnitude in mass.

Proof. Direct computation. Table 2 lists the predicted and observed masses. $\square$

Table 2: Mass-map fit. Predicted values use $A_{\text{mass}} = 224 \text{ GeV}$, $B_{\text{mass}} = 0.54$, and the $N/4$ labels from Table 1. Observed values are PDG central values (charged fermions in the $\overline{\text{MS}}$ scheme at 2 GeV where applicable).

Particle	$N/4$	Predicted (GeV)	Observed (GeV)
ν	60	1.89×10^{-12}	$< 10^{-9}$
e	24	5.27×10^{-4}	0.511×10^{-3}
u	21	2.67×10^{-3}	$\sim 2.2 \times 10^{-3}$
d	20	4.58×10^{-3}	$\sim 4.7 \times 10^{-3}$
s	14	0.116	~ 0.095
μ	14	0.116	0.106
c	9.5	1.33	1.27
τ	9	1.73	1.777
b	7.5	3.90	4.18
W	2	76.1	80.4
Z	2	76.1	91.2
H	1	130.5	125.1
t	0.5	171.0	172.7

The RMS log-ratio is 0.045 dex. This is the accuracy of the mass map as a fit; it is not a derivation of the mass values.

Remark 4.9. The mass map's coefficient $B_{\text{mass}} = (d-1)kL^2/(2N^2)$ contains the warp factor $kL = \ln(M_P/v_{\text{EW}})$, whose bare value $kL_{\text{bare}} = (d-1)N/(d+1) = 192/5 = 38.4$ is derived in HCSM-51a [14].

5 Framework-constant expressions in the labels

The following $N/4$ values have simple expressions in framework constants:

Particle	$N/4$ value as framework constant
ν	$N - d = 60$
e	$d! = V/2 = 24$
u	$(d - 1)(2d - 1) = 21$
d	$N/2 - H/2 + 4 = 20$
s, μ	$\text{mult}_{\lambda=4} = 14$
dark	$(d - 1)d = 12$
c	$(d - 1)^2 + (d - 1)/2 = 9.5$
τ	$(d - 1)^2 = 9$
b	$H/2 - 1/2 = 7.5$
W, Z	$d/2 = 2$
H	$H - d^2 = 1$
t	$2/d = 0.5$

The observation that all 14 labels have simple expressions in framework constants is a nontrivial structural fact of the anchor. It is a theorem *given* the anchor, not a derivation of the anchor's numerical values.

6 Inputs and derived

6.1 Derived from the substrate

1. The substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ and the framework constants $d = 4$, $N = 64$, $L = 8$, $H = 16$, $V = 48$ (HCSM-01).
2. The 14 modes at $\lambda = 4$ and their coordinates (n_1, n_2) (HCSM-04).
3. The D_4 orbit decomposition $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ (HCSM-04, HCSM-05).
4. The support classes $\{64, 48, 32\}$ and inverse participation ratios $\{2/128, 3/128, 4/128\}$ (HCSM-04, HCSM-05).
5. The Shannon entropy classes $\{6.0, 5.5, 5.0\}$ bits (HCSM-04, HCSM-05).
6. The winding generator H_{wind} , its spectrum, and the winding class partition $\{A, B, C\}$ (HCSM-14).
7. The processor $T = I - P_C$ and the F/B split (HCSM-13).
8. The chiral reflection rs , its fixed locus $\{W, t\}$, and its anticommutation with H_{wind} (HCSM-26).
9. The zero state s and its sign $\sigma = \text{sign}(s)$ (HCSM-22).

6.2 Inputs (empirical anchors)

1. **Input 1 (SM content).** The 14 particles of one Standard Model generation.
2. **Input 2 (Physical identifications).** The assignment of each of the 14 modes to a specific Standard Model particle (Thesis 1.1).
3. **Input 3 (Mass labels).** The 14 $N/4$ values of Eq. (1).

6.3 What is not claimed

We do not claim to derive the numerical values of $N/4$ from the substrate. The framework’s thesis is that the 14 modes are the 14 particles; the $N/4$ values are the framework’s mass-ordering labels for those particles. The thesis is empirical, in exactly the sense in which the Standard Model’s Yukawa couplings are empirical.

We do not claim that the structural relations of Section 4 would hold for arbitrary mass labels. They hold for the specific labels of Eq. (1). This is the content of the falsifiability conditions in Section 7.

We do claim that the structural relations are nontrivial theorems given the anchor: they tie together the D_4 orbit decomposition, the winding class partition, the fermion/boson split, the zero-state sign σ , and the chiral reflection rs in ways that the Standard Model does not.

7 Falsifiability

Theorem 7.1 (Falsifiability of the thesis). *The identification thesis is falsifiable. Any of the following observations would falsify it:*

1. *A fourth generation of Standard Model particles.*
2. *A particle content of one generation not matching the 14-mode structure.*
3. *An observed mass ordering that violates the rs -pair ordering (Theorem 4.2).*
4. *An observed W or t mass such that $N/4(W) + N/4(t) \neq (d + 1)/2$ or $N/4(W) - N/4(t) \neq (d - 1)/2$.*
5. *An observed pattern of masses such that the $5 + 9$ split (Theorem 4.7) fails.*
6. *An observed mass spectrum not fitting the mass map (Theorem 4.8) at any acceptable RMS.*

Proof. Each condition is a direct consequence of the thesis and the structural relations. If any is observed, the identification of the 14 modes with the Standard Model generation is incorrect. $\square$

8 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-23_verification`. It constructs the 14 modes, their coordinates, their D_4 orbits, their winding classes, their F/B content, their zero-state sign, and their $N/4$ labels from HCSM-native definitions, and verifies:

1. The (n_1, σ) bijection (Theorem 4.1).
2. The rs -pair ordering (Theorem 4.2).
3. The T_{half} pair sums (Theorem 4.3).
4. The observer-pair $N/4$ values (Theorem 4.4).
5. The four-cycle $N/4$ sum (Theorem 4.5).
6. The $5 + 9$ split (Theorem 4.7).

7. The mass-map fit at RMS 0.045 dex (Theorem 4.8).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

9 Conclusion

This paper has stated the identification of the 14 modes at $\lambda = 4$ with one Standard Model generation as the framework’s empirical thesis, and has derived the structural relations that the $N/4$ mass-ordering labels satisfy given the thesis.

The substrate-derived content of the framework is the 14 modes, their coordinates, their D_4 orbit decomposition, their support and IPR classes, their Shannon entropy classes, the winding generator and its class partition, the processor T and its F/B split, the chiral reflection rs and its fixed locus, and the zero state s . All of these are theorems of the torus and none of them depends on the empirical identification.

The empirical anchor of the framework is the assignment of each of the 14 modes to a specific Standard Model particle, together with the $N/4$ mass labels. This anchor plays the same role as the Standard Model’s Yukawa-coupling sector: it is the framework’s empirical content, and it is falsifiable.

Given the anchor, the following are theorems of the framework: the (n_1, σ) bijection, the rs -pair ordering, the T_{half} pair sums, the observer pair’s $N/4$ sum and separation, the four-cycle Klein orbit $N/4$ sum, the $5 + 9$ split into Substrate Eigenmodes and Mesh Resonances, and the mass-map fit at RMS 0.045 dex.

The framework’s strength is that it derives the particle content and the structure of one Standard Model generation from the 8×8 torus with a substantially smaller empirical anchor set than the Standard Model itself: 14 particle assignments and 14 mass labels versus the Standard Model’s ~ 30 free parameters, including the Yukawa couplings, mixing angles, and CP-violating phase. The framework is therefore a structural theory of the same type as the Standard Model, with additional derived content.

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A Web Crawler Metadata

A.1 Dublin Core

- DC.title: HCSM-23: The Identification of the 14 Modes with One Standard Model Generation
- DC.creator: Preschutti, Stanley
- DC.creator.orcid: 0009-0004-5445-1744
- DC.date.issued: 2026-09
- DC.type: Text / Preprint / White Paper
- DC.format: application/pdf; text/x-latex
- DC.language: en
- DC.publisher: Entropia Research Institute
- DC.rights: All rights reserved
- DC.subject: Hodge Complex Standard Model; HCSM; identification thesis; $N/4$ labels; mass map; empirical anchor; structural relations; 8×8 torus
- DC.description: HCSM-23 states the identification of the 14 modes at $\lambda = 4$ with one Standard Model generation as the framework's empirical thesis, and derives the structural relations that the $N/4$ mass labels satisfy given the thesis.
- DC.identifier: HCSM-23-2026-09

A.2 BibTeX

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@article{PreschuttiHCSM23,  
  author = {Preschutti, Stanley},  
  title  = {{HCSM-23}: The Identification of the 14 Modes  
           with One Standard Model Generation},  
  journal = {HCSM White Paper Series},  
  year   = {2026},  
  month  = {September},  
  note   = {ORCID: 0009-0004-5445-1744},  
  keywords = {Hodge Complex Standard Model, HCSM,  
             identification thesis, N/4 labels, mass map,  
             empirical anchor, structural relations},  
}
```